

Econ 131  
Spring 2026  
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Problem Set 1 — SOLUTIONS

## 1. True/False Statements [30 points]

[5 points each]

- (a) Difference-in-differences yields the causal effect of interest if the treatment group and the control group have identical outcome levels before the treatment begins.

**FALSE.** The DiD estimator requires **parallel trends**, not identical levels. Parallel trends means the treatment and control groups would have experienced the same *change* in outcomes over time in the absence of treatment, even if their levels differ. The treatment group can have systematically higher or lower outcome levels than the control group in the pre-period, as long as both groups are trending at the same rate. The DiD estimator differences out any constant level differences between groups.

- (b) In January 2025, the Congressional Budget Office released its updated budget outlook ...True/False/Uncertain: If  $r_t$  and  $g_t$  stay constant forever at their 2035 levels, the government can run a permanent primary deficit while maintaining a stable debt to GDP level.

**Depends on whether**  $r_{2035} < g_{2035}$  **or**  $r_{2035} > g_{2035}$ .

From Table B-1:

$$r_{2035} = \frac{\text{2035 Net Interest}}{\text{2034 Debt Held By The Public}} = \frac{1,783}{49,556} \approx 3.60\%$$
$$g_{2035} = \frac{\text{2035 GDP} - \text{2034 GDP}}{\text{2034 GDP}} = \frac{43,936 - 42,330}{42,330} \approx 3.79\%$$

The debt-to-GDP ratio evolves according to:

$$\Delta \left( \frac{D_t}{Y_t} \right) \approx (r_t - g_t) \frac{D_t}{Y_t} - s_t$$

where  $s_t$  is the primary surplus as a share of GDP.

For the debt-to-GDP ratio to remain stable with a permanent primary deficit ( $s_t < 0$ ), we need  $r_t < g_t$ . When  $g > r$ , GDP grows faster than the debt service burden, so the government can run a primary deficit while maintaining stable debt-to-GDP. With the 2035 CBO projections,  $r_{2035} \approx 3.60\% < g_{2035} \approx 3.79\%$ , so the answer is **TRUE**: the government can sustain a permanent primary deficit.

- (c) Adult income differences across U.S. local areas and across U.S. racial and ethnic groups are explained by differences in how rich or poor their parents were.

**FALSE.** Research by Chetty, Hendren, and others using the Opportunity Atlas and related studies shows that large differences in adult outcomes persist across places and

racial/ethnic groups even after conditioning on parental income. For example, Black men earn substantially less than white men even when raised by families with similar incomes in the same neighborhoods. Similarly, some regions (e.g., parts of the Southeast) exhibit much lower upward mobility than others (e.g., the Great Plains) for children from families at the same income level. These patterns reflect factors beyond parental income, including neighborhood effects, discrimination, differences in school quality, social capital, and other structural and environmental factors.

- (d) Suppose that a government applies a 10% income tax and uses the proceeds to give everyone a \$10,000 cash transfer. This tax-and-transfer system is progressive.

**TRUE.** Although the 10% income tax alone is proportional (the same rate applies to all incomes), the flat \$10,000 universal transfer represents a larger share of income for low-income individuals than for high-income individuals. Combined, the system is progressive: someone earning \$50,000 pays \$5,000 in tax and receives \$10,000, for a net gain of \$5,000 (net tax rate of  $-10\%$ ). Someone earning \$200,000 pays \$20,000 and receives \$10,000, for a net loss of \$10,000 (net tax rate of  $+5\%$ ). The breakeven income is \$100,000 (where \$10,000 tax exactly equals the \$10,000 transfer). Because the average net tax rate rises with income, the system is progressive.

- (e) If a government places a \$1 per unit tax on producers of cigarettes rather than on consumers, producers will bear a larger economic burden of the tax than they would if the tax were placed on consumers instead.

**FALSE.** This confuses statutory incidence (who legally pays) with economic incidence (who actually bears the burden). A fundamental result in tax incidence theory is that the equilibrium outcome is *independent of which side of the market is legally required to pay the tax*. Whether the \$1 tax is nominally imposed on producers or consumers, the economic incidence will be the same. The tax creates a wedge between the price consumers pay and the price producers receive, and market forces determine how this burden is split based on relative elasticities, not based on who writes the check.

- (f) When a sales tax is made more visible and salient to consumers at the point of purchase, it breaks the standard tax incidence result that equilibrium is independent of whether the tax is nominally placed on consumers or producers.

**TRUE.** Empirical evidence (e.g., Chetty, Looney, and Kroft 2009) shows that tax salience matters. When sales taxes are displayed prominently (e.g., included in posted shelf prices rather than added at checkout), consumers respond more strongly. This increased salience effectively makes demand more elastic with respect to the tax, violating the standard model's assumption of full rationality. Since consumers respond differently to economically

equivalent taxes depending on their visibility, statutory design and presentation can affect economic outcomes.

## 2. Difference-in-Differences (DiD) and Parallel Trends [30 points]

### Scenario 1 [18 points]

- (a) [4 points] Compute the four group means.

Control group:

$$\begin{aligned}\bar{Y}_{C,\text{Pre}} &= \frac{50 + 51 + 52 + 53 + 54}{5} = \frac{260}{5} = 52 \\ \bar{Y}_{C,\text{Post}} &= \frac{55 + 56 + 57 + 58 + 59 + 60}{6} = \frac{345}{6} = 57.5\end{aligned}$$

Treated group:

$$\begin{aligned}\bar{Y}_{T,\text{Pre}} &= \frac{55 + 56 + 57 + 58 + 59}{5} = \frac{285}{5} = 57 \\ \bar{Y}_{T,\text{Post}} &= \frac{62 + 63 + 64 + 65 + 66 + 67}{6} = \frac{387}{6} = 64.5\end{aligned}$$

- (b) [4 points] Compute changes within each group.

$$\Delta_C = 57.5 - 52 = 5.5$$

$$\Delta_T = 64.5 - 57 = 7.5$$

- (c) [5 points] Compute the DiD estimate and interpret.

$$\hat{\tau}_{DiD} = \Delta_T - \Delta_C = 7.5 - 5.5 = 2.0$$

The DiD estimate of 2.0 indicates that the policy caused the outcome to increase by 2 units on average for the treated group, after accounting for the general time trend observed in the control group.

- (d) [5 points] Parallel trends check.

Students should plot both time series. Pre-2015, both lines increase with a slope of +1 per year. The treated group is consistently 5 units higher than the control group, but both trend upward at the same rate. At 2015, the treated group jumps from 59 to 62, while the control group continues its trend from 54 to 55. Post-2015, both groups continue with slope +1 per year, but the gap widens to 7 units.

**Yes**, parallel trends looks plausible. Both groups show identical linear trends (slope of +1 per year) during 2010–2014. The only difference is a constant level shift of 5 units, which DiD accounts for.

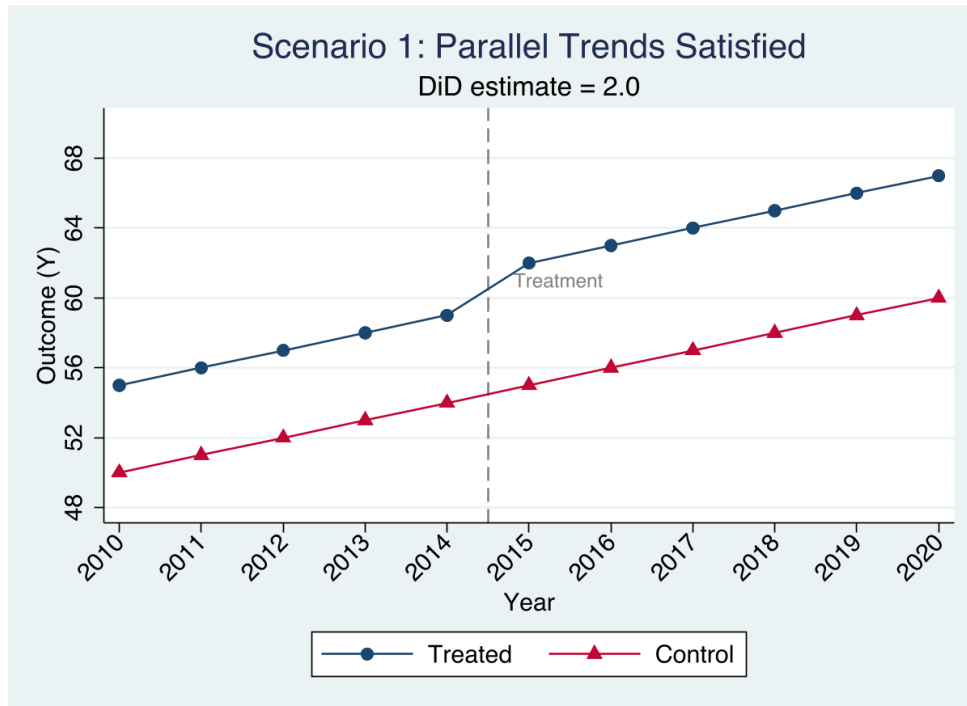


Figure 1: Scenario 1: Parallel trends satisfied

## Scenario 2 [12 points]

(e) [5 points] Repeat the same steps.

Control group means: same as Scenario 1 ( $\bar{Y}_{C,Pre} = 52$ ,  $\bar{Y}_{C,Post} = 57.5$ ).

Treated group:

$$\bar{Y}_{T,Pre} = \frac{53 + 55 + 57 + 59 + 61}{5} = \frac{285}{5} = 57$$

$$\bar{Y}_{T,Post} = \frac{66 + 66 + 65 + 64 + 63 + 63}{6} = \frac{387}{6} = 64.5$$

Changes:  $\Delta_C = 5.5$ ,  $\Delta_T = 7.5$ .

$\hat{\tau}_{DiD} = 7.5 - 5.5 = 2.0$  (identical to Scenario 1).

(f) [4 points] Plot and diagnose parallel trends.

Pre-2015: the control group increases by +1 per year, while the treated group increases by +2 per year. These are *not* parallel. Post-2015: the treated group decreases by about 0.6 per year on average ( $66 \rightarrow 63$ ), while the control group continues increasing.

**No**, parallel trends does not look plausible. The treated group was trending upward twice as fast as the control group before treatment (slope +2 vs. +1). The post-treatment trend reversal is also suspicious and suggests mean reversion.

(g) [3 points] Why is Scenario 2 more concerning?

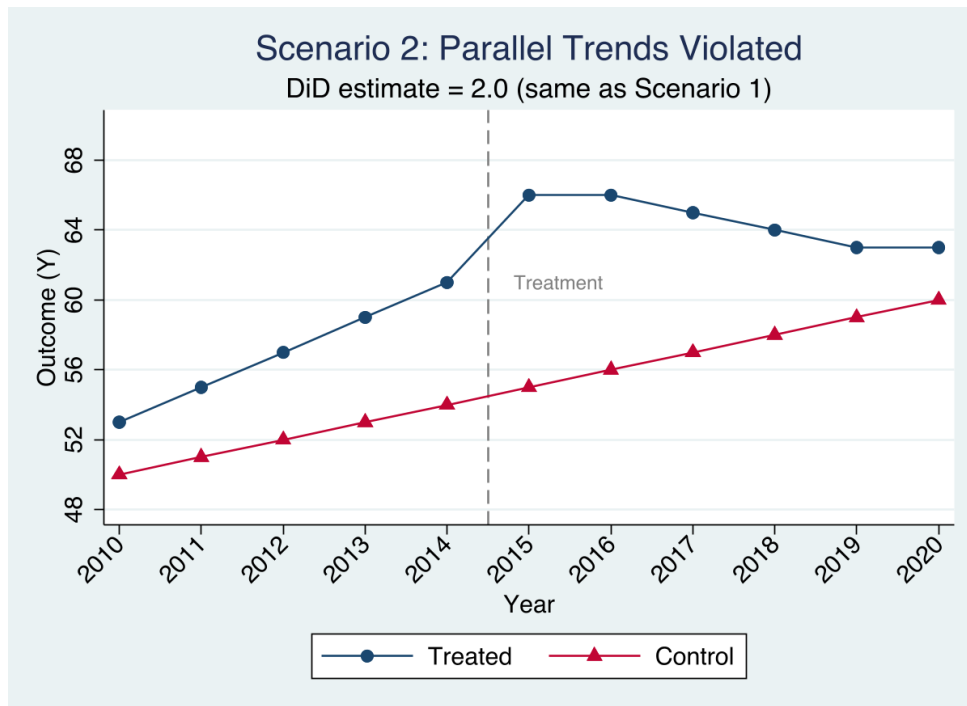


Figure 2: Scenario 2: Parallel trends violated

Even though the DiD estimate is identical (2.0), Scenario 2 violates the parallel trends assumption because the treated group was already trending upward faster than the control group before treatment. The DiD estimate likely overstates the true treatment effect and may simply reflect continuation of pre-existing differential trends rather than a causal effect of the policy.

### 3. Tariffs and Welfare [28 points]

- (a) [2 points] Circle and label the world price pre-tariff, the tariff-inclusive domestic price, and draw a vertical bracket showing the tariff wedge.

The world price pre-tariff ( $P_W = \$300$ ) is the horizontal line at the lower price level. The tariff-inclusive domestic price ( $P_T = P_W + t = \$350$ ) is the horizontal line at the higher price level. The tariff wedge is the vertical distance between  $P_W$  and  $P_T$  which is \$50.

- (b) [3 points] Label domestic quantity supplied and demanded before and after the tariff, and mark import quantities.

Before the tariff: domestic supply  $S_1$  (where supply intersects  $P_W$ ) and domestic demand  $D_1$  (where demand intersects  $P_W$ ). Imports before =  $D_1 - S_1$ .

After the tariff: domestic supply  $S_2$  (where supply intersects  $P_T$ ) and domestic demand  $D_2$  (where demand intersects  $P_T$ ). Imports after =  $D_2 - S_2$  (smaller than before).

The tariff reduces imports because domestic supply increases and domestic demand decreases at the higher price.

- (c) [8 points] Interpret the four numbered areas. [2 points per area]

- Area 1: Producer surplus gain. Goes to domestic producers. Gain for producers, transfer from consumers.
- Area 2: Production distortion / deadweight loss. Goes to no one. Loss for the home country. Represents inefficiency from producing domestically at costs above the world price.
- Area 3: Government tariff revenue ( $= t \times M_2$ ). Goes to the government. Transfer from consumers.
- Area 4: Consumption distortion / deadweight loss. Goes to no one. Loss for the home country. Represents lost consumer surplus from forgone purchases valued above the world price.

Total consumer surplus loss = Areas 1 + 2 + 3 + 4. Of this, Areas 1 and 3 are transfers (to producers and government), while Areas 2 and 4 are pure deadweight losses.

- (d) [3 points] Identify the deadweight loss triangles and explain the distortion each captures.

The deadweight loss triangles are Area 2 and Area 4.

Area 2 (production distortion): efficiency loss from producing too much domestically. Units between  $S_1$  and  $S_2$  cost more to produce than the world price, so they should have been imported.

Area 4 (consumption distortion): efficiency loss from consuming too little. Consumers valued units between  $D_2$  and  $D_1$  above the world price but forgo them because of the tariff-inflated domestic price.

(e) [8 points] Analytical tariff calculation.

Given:  $Q^D = 100 - P$ ,  $Q^S = P$ ,  $P_W = 20$ ,  $t = 10$ .

i. [2 points] Without a tariff:

At  $P_W = 20$ :

$$Q^D = 100 - 20 = 80 \text{ thousand tons}$$

$$Q^S = 20 \text{ thousand tons}$$

$$\text{Imports} = 80 - 20 = 60 \text{ thousand tons}$$

ii. [2 points] With a tariff of  $t = 10$ :

Domestic price:  $P_T = P_W + t = 20 + 10 = 30$ .

At  $P_T = 30$ :

$$Q^D = 100 - 30 = 70 \text{ thousand tons}$$

$$Q^S = 30 \text{ thousand tons}$$

$$\text{Imports} = 70 - 30 = 40 \text{ thousand tons}$$

iii. [2 points] Welfare effects:

Consumer surplus loss:

$$\text{CS before} = \frac{1}{2}(100 - 20)(80) = 3,200$$

$$\text{CS after} = \frac{1}{2}(100 - 30)(70) = 2,450$$

$$\text{CS loss} = 3,200 - 2,450 = 750$$

Producer surplus gain:

$$\text{PS before} = \frac{1}{2}(20)(20) = 200$$

$$\text{PS after} = \frac{1}{2}(30)(30) = 450$$

$$\text{PS gain} = 450 - 200 = 250$$

Government tariff revenue:

$$\text{Revenue} = t \times \text{Imports} = 10 \times 40 = 400$$

Total deadweight loss:

$$\text{DWL} = \text{CS loss} - \text{PS gain} - \text{Revenue} = 750 - 250 - 400 = 100$$

iv. [2 points] Verify using distortion triangles:

$$\text{Production distortion: } \frac{1}{2}(30 - 20)(30 - 20) = \frac{1}{2}(10)(10) = 50$$

$$\text{Consumption distortion: } \frac{1}{2}(80 - 70)(30 - 20) = \frac{1}{2}(10)(10) = 50$$

$$\text{Total: } 50 + 50 = 100 \quad \checkmark$$

(f) [4 points] How would the analysis differ for a large country?

When a large country imposes a tariff, its reduced import demand is substantial enough to lower the world price. This creates a terms-of-trade gain: the large country pays less for the imports it continues to purchase, extracting surplus from foreign producers. The overall welfare impact becomes ambiguous because this terms-of-trade gain may partially or fully offset the deadweight losses. If the tariff is set at the “optimal” level, the terms-of-trade gain can exceed the deadweight losses, making the large country better off. However, this is a “beggar-thy-neighbor” policy, and if trading partners retaliate, all countries can end up worse off than under free trade.

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*End of Solutions*